

AI in Modern Warfare: Balancing Innovation with Ethical Integrity – Opportunities and Challenges

Student Name: Mohammad Aabid Hasan
Student ID: 2463016
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Abstract

This report looks at how artificial intelligence (AI) is used in modern warfare. It shows the good things AI can do, like making weapons, watching things, and helping decisions faster and better. But it also talks about big ethical problems.

The main purpose is to see how AI helps the military do things more quickly and accurately, which can save soldiers' lives. However, there are serious risks. For example, machines can make decisions to kill without humans fully controlling them, which makes it hard to know who is responsible if something goes wrong. Also, AI in surveillance can watch too many people and hurt privacy, and it might make wars start or get worse by mistake.

The key ethical worries are: letting machines decide who lives or dies, keeping national security strong without taking away people's freedom and rights, and making sure the harm is not too much.

From this study, we see that AI can bring good changes by making operations more precise and reducing human danger. But we really need strong rules and frameworks to keep things ethical. I suggest a simple ethical AI framework with three main parts: fairness (no bias in decisions), transparency (people can understand how AI works), and human oversight (humans must always have control over important choices).

In the end, if countries follow good international guidelines, using AI in warfare can help make progress fair and safe for everyone

Introduction

Ethics in AI is very important because these technologies are now used in key areas, including warfare, where decisions can change or end lives. Modern AI systems create big ethical, social, and moral problems. For example, biased algorithms can continue

discrimination, privacy gets lost from too much data collection, and human responsibility becomes weak when machines decide alone. In the military, these issues become even more serious: AI in autonomous weapons might attack civilians by mistake due to bad data, and surveillance tools can take away people's freedoms while claiming it is for security. Socially, unequal access to AI creates unfair power differences between nations and raises moral questions about fairness.

Many ethical principles, guidelines, and rules exist to solve these problems. The OECD AI Principles support values like robustness, accountability, and respect for human-centered values. The EU AI Act groups AI systems by risk levels and requires strong checks for high-risk uses (though it mostly does not cover pure military applications). These frameworks help ensure AI follows society's rules and international laws.

Ethical and responsible AI in warfare is extremely important. As AI develops, it offers chances for more precise and efficient work, but it can also increase harms if not controlled. This report shows the need for balanced innovation and explains how ethical rules can guide sustainable progress in areas like governance and security.

Thematic Review

The use of artificial intelligence (AI) in modern warfare is a big change. It brings new speed, better work, and fresh ideas, but it also creates serious ethical problems.

In autonomous weapon systems (AWS), AI helps find and attack targets very fast. This can keep soldiers safer and reduce mistakes that hurt innocent people because of precise work. But big ethical questions come when machines decide who to kill. This makes it hard to follow International Humanitarian Law (IHL) and decide who is responsible if the AI makes a wrong choice – is it the programmer, the person using it, or the country that sent it? Other problems are bias in the algorithm that can attack some groups more than others, lack of transparency because "black box" systems hide how decisions are made, and unfairness if the AI learns mostly from data of powerful countries (Horowitz & Michael, 2016). On the good side, AI can help society by using

predictive analytics to see possible conflicts early and allow countries to talk and stop problems before they start.

In surveillance and intelligence, AI makes it easier to look at data from drones and satellites. This improves finding threats and makes national security stronger. But it also creates risks like too much watching of people, which hurts privacy and civil rights because large amounts of data can be used to monitor civilians without reason (Sophie, Zureck, & Alexander, 2025). To balance security and people's rights, we need ethical rules like collecting only necessary data and getting proper permission. On a bigger level, if only rich or advanced countries have good AI surveillance, it can make the world more unequal and unfair. But AI also gives chances for good things, like using it to send help quickly in wars or disasters, which helps more people.

AI can also make conflicts worse by accident, for example through wrong calculations in automatic systems or when countries hurry to build and test new weapons, which can bring the world closer to nuclear danger (Payne & Kenneth, 2024). Nations and developers have the duty to make sure actions are fair and do not cause extra harm. They must work on transparency so no hidden decisions cause big problems. Other challenges include losing jobs in military fields because of AI and unfair access to AI in areas like education and finance. To get the best results, we need ideas like team reviews with ethics experts and working together between countries.

Many national and international efforts support ethical AI. UNESCO's guidelines push for AI that puts humans first, with focus on sustainability and fairness. OECD principles talk about inclusive growth and accountability (Development, 2023). The EU AI Act controls high-risk military AI with required checks. NIST's RMF gives practical tools to manage risks, which can be used in warfare too (Technology, 2023). Other rules, like the U.S. DoD Ethical Principles, also say responsible use is important (Williams & Heather, 2021).

I suggest a simple ethical AI framework for warfare with these main parts:

- **Fairness:** Use different training data so algorithms are not biased.

- **Transparency:** Make AI explainable so people can check how it works.
- **Human Oversight:** Always let humans control big decisions, especially killing.
- **Sustainability:** Build systems that do not harm the environment or society in the long run.
- **Accountability:** Make clear rules so someone is responsible for what AI does.

This framework can be used by testing again and again and listening to different people. It can help make AI ethical not just in war, but in many other fields too.

Discussion / Personal Reflection

When I think about this research, I feel that ethical AI in warfare is very important. It protects human dignity even when technology becomes very powerful. If we add ethics from the start, AI will not become a tool for too much power without control. This way, the effects on society can stay good and not create divisions.

From studying autonomous weapons, I learned that AI can be very accurate and save lives, but the big problem is who is responsible if it makes a mistake. This gap can make people lose trust in governments and institutions. It shows how hard it is to mix new technology with moral rules (Technology, 2023). Another important thing I understood is that AI has two sides – it can help security a lot, but it can also hurt privacy badly if not controlled properly. This makes us see why we need good, balanced rules.

Following ethical principles can make AI much safer and fair for everyone. It reduces bias and helps include all kinds of people. For example, keeping humans in control stops wars from getting bigger by mistake and supports better global cooperation. In the long run, ethical AI can bring peace and better economy, but if we use it without ethics, it can increase unfairness, more arms races, or even very big dangers.

This study made me believe strongly that AI should help humans, not take their place. Developers must always put human values first to build a fair and just world.

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